



# Portfolio Svitlana Basurina

## 1. Project Planning

### Project 1: Sales Performance Dashboard

- **Goal:** Show KPIs for revenue, profit, and top products.
- **Skills shown:** Power BI visuals, DAX, star schema.
- **Dataset:** Superstore

### Project 2: SQL Business Analysis + Intelligence

- **Goal:** Query orders/customers to find trends, top customers, repeat rate.
- **Skills shown:** SQL queries, data cleaning, joins, DAX.
- **Dataset:** E-commerce Orders

### Project 3: DAX & Advanced Analysis

- **Goal:** Build measures with time intelligence (YoY growth, rolling revenue).
- **Skills shown:** DAX measures, business insights.
- **Dataset:** Either Superstore or combine with SQL project

## Project 1: Prepare Superstore Dataset

Superstore.

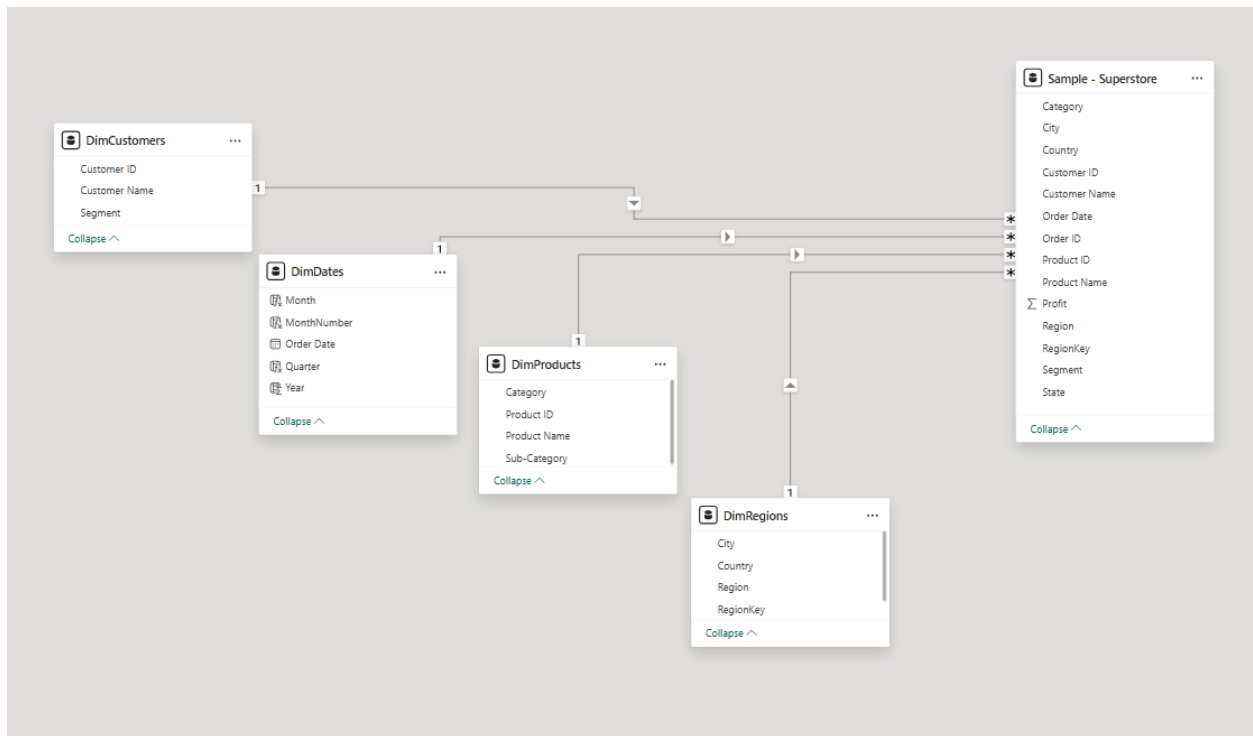
### Tools & Skills Used

- Power BI Desktop
- Power Query (ETL)

- DAX (calculated columns & measures)
- Data Modeling (Star Schema)
- Data Visualization & Dashboard Design

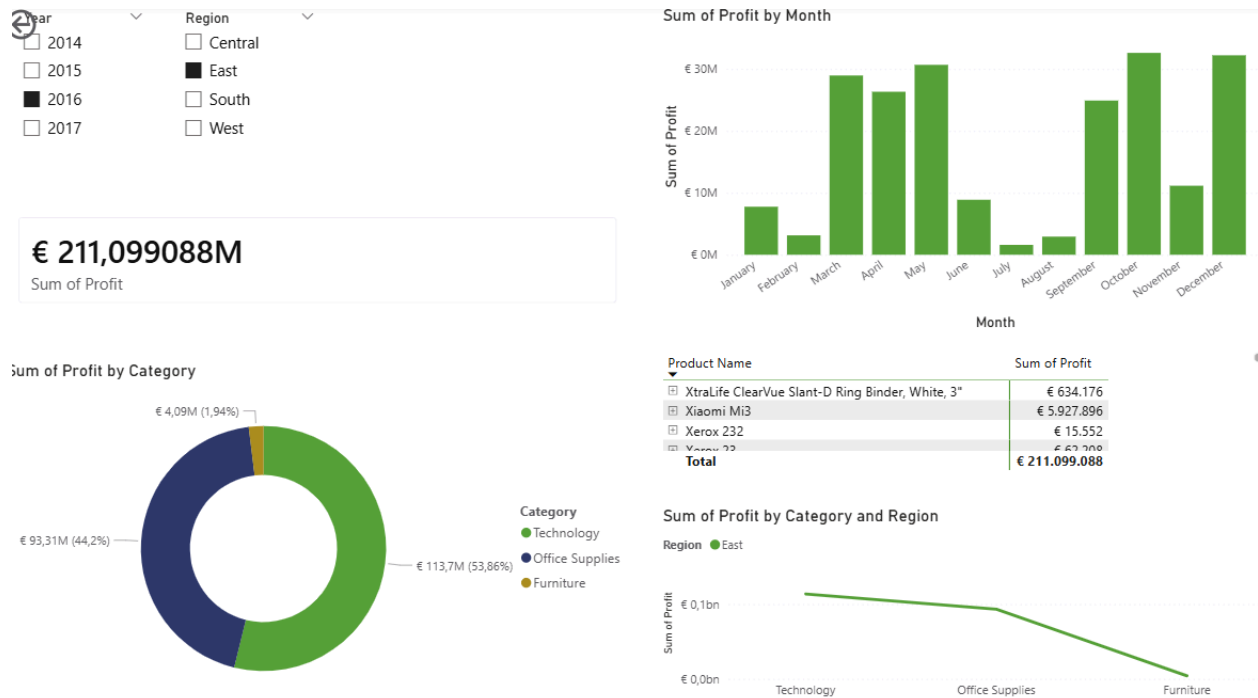
## Tasks:

1. Explore in Power BI
2. Check:
  - Missing values
  - Duplicate rows
  - Correct data types (US Dates ⇒ European)
3. Plan **star schema** for Power BI:
  - Fact Table: Sales (Order ID, Revenue, Quantity, Profit)
  - Dimension Tables: Customers, Products, Regions, Dates
4. ProductID appears multiple times in the fact table by design, as each row represents a transaction. Uniqueness is enforced in the Products dimension to enable a proper star schema relationship.

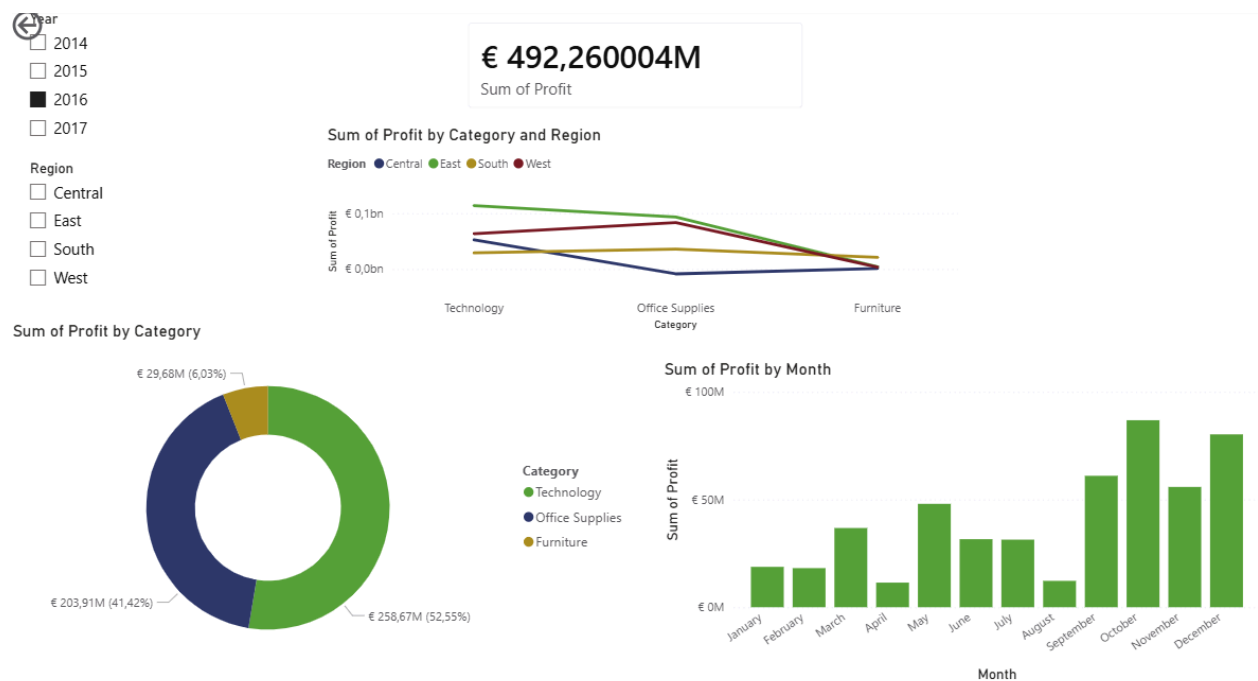


5. Year = YEAR('Date'[Date])  
Month = FORMAT('Date'[Date], "MMMM")  
Quarter = "Q" & FORMAT('Date'[Date], "Q")
6. Enhanced the Dates dimension by adding Year, Month, and Quarter calculated columns in DAX to enable intuitive time-based slicing and reporting
7. Created visualizations:

Version 1.



Version 2.



Designed an interactive sales dashboard in Power BI using a star schema model. Implemented time, geography, and product slicers to enable dynamic analysis of

revenue, profit, and quantity sold across regions and periods.

## Key Insights Enabled

- Identify high-performing regions and products
- Track sales trends over time
- Analyze performance interactively by multiple dimensions

## Project 2: SQL Business Analysis

### Project Goal

Answer **business questions** using SQL:

- Who are the top customers?
- How often do customers repeat purchases?
- What trends exist over time?

### E-Commerce Database.

1. Imported raw e-commerce CSV into staging table, cleaned invalid dates and duplicate order IDs, and loaded into final table with proper types. SQL Server queries were then run for analysis.
  - Dropping/creating staging table
  - Importing CSV
  - Cleaning invalid dates
  - Removing duplicates
  - Producing final `EcommerceData` table
2. Encountered data integrity issues where trailing Carriage Return characters ( `CHAR(13)` ) caused filtering failures; resolved by implementing a SQL cleaning script to strip non-printing control characters and standardizing country names.
  - **Ingested** a messy CSV with conflicting line endings.
  - **Repaired** 4,796 rows where data had "slid" across columns.

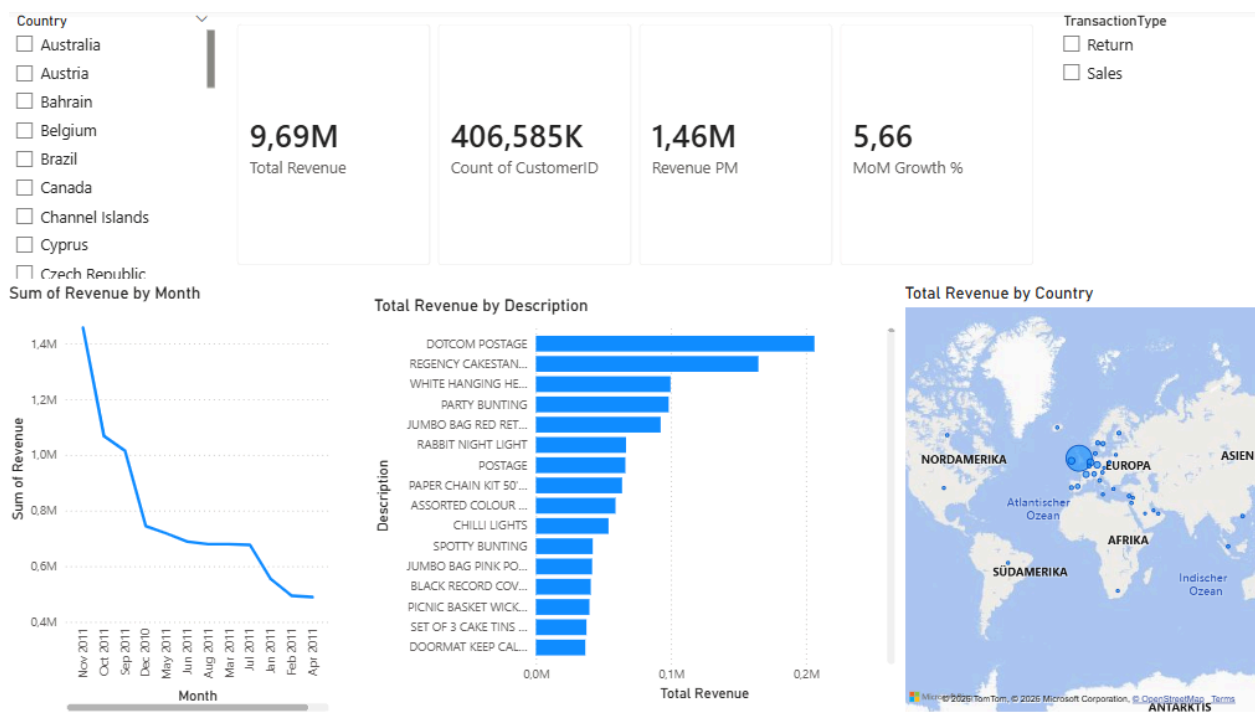
- **Identified and Nuked** hidden `CHAR(13)` carriage returns.
- **Validated** a \$29M dataset with a 100% accuracy match between calculated and stored revenue.

## 2. Analyzed `EcommerceData` table:

Ran analysis queries for:

- Top 10 customers by revenue
- Monthly revenue trends
- Repeat customers
- Top products by revenue

## 3. Connected SQL Server to PowerBI and vizualized results:



Developed a full-stack BI solution using SQL Server and Power BI. Engineered an ETL pipeline to recover ~4,800 corrupted rows and standardize international sales data. Designed a star-schema data model and implemented DAX Time Intelligence measures to track Month-over-Month growth across 37 countries.

Here is a summary of the key milestones achieved:

- **Database Architecture & Ingestion:** Established a dedicated SQL environment ( `Ecommerce_Analysis` ) and implemented a two-stage ETL process using **Staging Tables** to safely import 500k+ rows of raw CSV data using `BULK INSERT` .
- **Complex Data Recovery:** Identified and resolved a critical data integrity issue where ~4,800 rows were "corrupted" (data shifted across columns). Developed a custom **string manipulation script** using `CHARINDEX` and `REVERSE` to surgically extract and re-align Price, Customer ID, and Country values.
- **Advanced Data Cleaning:** Programmatically removed hidden non-printing characters-specifically **Carriage Returns ( `CHAR(13)` )** and **Line Feeds ( `CHAR(10)` )**- that were causing filtering failures and incorrect grouping in the Country column.
- **Performance Optimization:** Built a refined production table ( `OrdersWithRevenue` ) using **Persisted Computed Columns** to handle real-time revenue calculations (Quantity×UnitPrice) without sacrificing query speed.
- **Relational Data Modeling:** Transitioned from a flat-file structure to a **Star Schema** in Power BI. Developed a gapless **DAX Calendar Table** and established a 1-to-Many relationship between the date dimension and transaction facts.
- **Business Intelligence & Analytics:** Authored complex **DAX measures** for financial reporting, including:
  - **Total Revenue:** Dynamic summation of global sales.
  - **Time Intelligence:** Created "Prior Month" and **Month-over-Month (MoM) Growth %** metrics to track business momentum.
- **Data Visualization:** Designed an interactive dashboard featuring Geographic Map visuals, Sales Trend line charts, and **Conditional Formatting** to highlight performance shifts.

## ⇒ Business Intelligence:

Total revenue is **9.69M**, but looking at the map, the vast majority is concentrated in Northern Europe (the UK). This indicates a highly **localized** success story rather than a global one.

Top product by revenue is "**DOTCOM POSTAGE**". It means shipping fees are a significant portion of the "sales." If customers are paying that much for postage, they might switch to a competitor with lower shipping costs or better local distribution.

Since the dataset is sorted chronologically, it shows the business is losing momentum as the year progresses.

**Average Revenue Per User (ARPU)** is roughly **\$23.83**. This suggests a high-volume, low-margin "gift shop" style business.